Day 7 — CSS Fundamentals

Simple Questions, Answers, CSS Examples, and Responsive Webpage Practice

1. What is CSS?

CSS stands for Cascading Style Sheets. It controls the appearance of HTML pages, including colours, fonts, spacing, layout, and how content adapts to different screen sizes.

2. What are inline, internal, and external CSS?

Inline CSS is written in an element's style attribute. Internal CSS is written inside a style element in the HTML head. External CSS is stored in a separate .css file and linked to the HTML page. External CSS is generally best for reusable styles across a website.

```
<!-- Inline CSS -->
<p style="color: blue;">Hello</p>

<!-- Internal CSS, inside <head> -->
<style>
  p { color: blue; }
</style>

<!-- External CSS, inside <head> -->
<link rel="stylesheet" href="style.css">
```

3. What are CSS selectors and properties?

A selector chooses which HTML elements to style. A property describes what to change, and a value sets the desired result. Common selectors include element selectors, .class selectors, #id selectors, and descendant selectors.

```
p { color: #333333; }       /* element */
.card { padding: 16px; }   /* class */
#main-title { font-size: 2rem; } /* ID */
```

4. How do colours and backgrounds work?

Use color to set text colour and background-color to set an element's background. Colours can be written as names, hexadecimal values, RGB, or HSL.

```
body {
  color: #222222;
  background-color: #f4f7fb;
}
```

5. What are margin and padding?

Margin is the space outside an element's border. Padding is the space between the content and its border.

```
.card {
  margin: 20px;   /* outside */
  padding: 16px;  /* inside */
  border: 1px solid #dddddd;
}
```

6. What is the CSS box model?

The box model consists of content, padding, border, and margin. Using box-sizing: border-box makes an element's declared width include its padding and border, which often makes layouts easier to manage.

```
* {
  box-sizing: border-box;
}
```

7. What is Flexbox?

Flexbox is a CSS layout system for arranging items in a row or column. The parent becomes a flex container when display: flex is applied. Common properties include justify-content, align-items, gap, and flex-wrap.

```
.cards {
  display: flex;
  justify-content: center;
  align-items: stretch;
  gap: 16px;
  flex-wrap: wrap;
}
```

8. What is responsive design?

Responsive design lets a webpage work well on mobile, tablet, and desktop screens. Use flexible widths, responsive images, Flexbox or Grid, and media queries. The viewport meta tag helps mobile browsers display the page at the device width.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">

@media (max-width: 600px) {
  .navigation {
    flex-direction: column;
  }
  .cards {
    flex-direction: column;
  }
}
```

9. Practice Task: Responsive Personal Profile Page

Save the following as profile.html and style.css in the same folder. Replace sample details with your own.

```
<!-- profile.html -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Profile</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="site-header">
    <h1>Your Name</h1>
    <p>Student | Aspiring Developer</p>
  </header>

  <nav class="navigation">
    <a href="#about">About</a>
    <a href="#skills">Skills</a>
    <a href="#contact">Contact</a>
  </nav>

  <main class="container">
    <section class="card" id="about">
      <h2>About Me</h2>
      <p>I am learning HTML and CSS.</p>
    </section>

    <section class="card" id="skills">
      <h2>My Skills</h2>
      <ul>
        <li>HTML</li><li>CSS</li><li>Java</li>
      </ul>
    </section>

    <section class="card" id="contact">
      <h2>Contact</h2>
      <p>student@example.com</p>
    </section>
  </main>

  <footer class="site-footer">
    <p>My Profile Page</p>
  </footer>
</body>
</html>
```

```
/* style.css */
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  line-height: 1.6;
  color: #222;
  background-color: #f4f7fb;
}

.site-header {
  padding: 28px 16px;
  text-align: center;
  color: white;
  background-color: #17365d;
}

.navigation {
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
  gap: 20px;
  padding: 14px;
  background-color: #244a73;
}

.navigation a {
  color: white;
  text-decoration: none;
}

.navigation a:hover {
  text-decoration: underline;
}

.container {
  display: flex;
  flex-wrap: wrap;
  gap: 16px;
  max-width: 1000px;
  margin: 24px auto;
  padding: 0 16px;
}

.card {
  flex: 1 1 250px;
  padding: 18px;
```

```
  background: white;
  border: 1px solid #dce3eb;
  border-radius: 8px;
}

.site-footer {
  padding: 12px;
  text-align: center;
}

@media (max-width: 600px) {
  .navigation {
    flex-direction: column;
    align-items: center;
    gap: 10px;
  }
  .container {
    flex-direction: column;
  }
}
```

10. Practice Task: Responsive Student Registration Form

This example improves spacing and layout on mobile and desktop. HTML forms need a backend to store submitted data; this example demonstrates the frontend only.

```
<!-- registration.html -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Registration</title>
  <link rel="stylesheet" href="form.css">
</head>
<body>
  <main class="form-card">
    <h1>Student Registration</h1>
    <form action="#" method="post">
      <label for="name">Full Name</label>
      <input id="name" name="name" type="text" required>

      <label for="email">Email</label>
      <input id="email" name="email" type="email" required>

      <label for="age">Age</label>
      <input id="age" name="age" type="number" min="1" max="120">

      <label for="course">Course</label>
      <select id="course" name="course">
        <option value="bsc">BSc</option>
        <option value="mca">MCA</option>
        <option value="other">Other</option>
      </select>

      <button type="submit">Register</button>
    </form>
  </main>
</body>
</html>
```

```
/* form.css */
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  padding: 24px 12px;
  font-family: Arial, sans-serif;
  background: #f4f7fb;
}

.form-card {
  width: 100%;
  max-width: 520px;
  margin: 0 auto;
  padding: 24px;
  background: white;
  border-radius: 10px;
  border: 1px solid #dce3eb;
}

label {
  display: block;
  margin-top: 14px;
  font-weight: bold;
}

input, select {
  width: 100%;
  padding: 11px;
  margin-top: 5px;
  border: 1px solid #aab7c4;
  border-radius: 5px;
  font: inherit;
}
```

```
button {
  width: 100%;
  margin-top: 20px;
  padding: 12px;
  border: 0;
  border-radius: 5px;
  background: #17365d;
  color: white;
  font: inherit;
  cursor: pointer;
}

@media (max-width: 480px) {
  .form-card {
    padding: 16px;
  }
}
```

Day 7 — Task Checklist

■ Add an external CSS file to the personal profile page.

■ Style headings, text, colours, backgrounds, margins, and padding.

■ Create a navigation bar using Flexbox.

■ Arrange profile cards using Flexbox and gap.

■ Improve the registration form with labels, spacing, and full-width inputs.

■ Test both pages at desktop and mobile widths.

Quick Revision

CSS: styles HTML. **Selector:** chooses elements. **Margin:** outside space. **Padding:** inside space. **Box model:** content + padding + border + margin. **Flexbox:** arranges items in a row or column. **Media query:** applies styles based on screen conditions.